COSC 2633/2637
Big Data Processing

Week 1
Introduction to Big Data Processing

RMIT UNIVERSITY

Acknowledgement of country

RMIT University acknowledges the Wurundjeri people of the Kulin Nations as the Traditional Owners of the land on which the University stands. RMIT University respectfully recognises Elders past, present and future.

Part 1

Course Administration

Course administration

Prerequisite Knowledge

- This prerequisite knowledge – Undergraduate

COSC2803 - Java Programming Studio
OR
COSC3106 - Python Programming Studio

- This prerequisite knowledge – Postgraduate

COSC1295 Advanced Programming
OR
COSC2820/COSC3015 Advanced Programming for Data Science
AND
ISYS1055 Database Concepts

Course administration

Lectorial and Practical Classes

This course will be delivered face-to-face and by accessing materials and resources on Canvas.

- **Lecture Notes and Recordings**
  - Study weekly lecture notes and recordings before interactive sessions
  - Complete the prescribed reading if they are specified for the week
- **Interactive Learning Activities**
  - Lectorial Sessions (starting from week 1)
    2 hours per week via Collaborate Ultra - recorded
  - Laboratory sessions (starting in week 2)
    2 hours per week via Collaborate Ultra – recorded

Communication with Teachers

- Face-to-face consultation (time and location can be found Canvas -> Syllabus)
- Online consultation via Teams (pre-booking by RMIT email is needed)
- By posting question in discussion forum

Course administration

Canvas

This semester we are using the Canvas system accessible through myRMIT which will include access to online resources including:

- Syllabus – course timetable
- Modules – weekly learning materials
- Assignments – assessment tasks
- Collaborate Ultra – online lectorials and practical classes
- Discussion forums – Q&A
- Grade – assignment results

Course administration

Learning Materials

No compulsory textbook

All Learning materials are available on Canvas

- Weekly modules
  - Lecture notes and recordings
  - Practical class learning materials
- Collaborate Ultra
  - Lectorial session recordings
  - Practical class recordings
- Readings
  - *Data-Intensive Text Processing with MapReduce by Jimmy Lin and Chris Dyer, University of Maryland, College Park, Manuscript prepared April 11, 2010*
  - *Other specified readings*

End of Part 1

Part 2

What is Big Data?

Key big data statistics for 2026

- **How fast is data still growing?** Global mobile data traffic is now exceeding **210 exabytes (EB) per month**. Driven primarily by video streaming (which accounts for ~75% of traffic) and the rapid growth of 5G users, global mobile network data traffic is projected to reach over **500 exabytes per month by 2031**

Decimal			Binary				
Value		Metric	Value		IEC		Legacy
1000	kB	kilobyte	1024	KiB	kibibyte	KB	kilobyte
1000^2	MB	megabyte	1024^2	MiB	mebibyte	MB	megabyte
1000^3	GB	gigabyte	1024^3	GiB	gibibyte	GB	gigabyte
1000^4	TB	terabyte	1024^4	TiB	tebibyte	TB	terabyte
1000^5	PB	petabyte	1024^5	PiB	pebibyte		–
1000^6	EB	exabyte	1024^6	EiB	exbibyte		–
1000^7	ZB	zettabyte	1024^7	ZiB	zebibyte		–
1000^8	YB	yottabyte	1024^8	YiB	yobibyte		–

Key big data statistics for 2026

- **How big is the big data analytics market?** It reaches **$447.68 billion in 2026** and is on course to hit $1.18 trillion by 2034.
  - Market Valuations & Forecasts: Review the detailed breakdown of the Itransition Future of Big Data Report to see exactly how software and services are driving the trillion-dollar projection.
  - Global Statistics: Check the G2 Big Data Statistics Overview for a deep dive into segment shares and regional growth trends.

Key big data statistics for 2026

- **How much of that data actually gets used?**
  - Dark data is the information that organizations accumulate but often never use for analytics or decision-making.
  - Most companies today store vast quantities of dark data. In Splunk's global research survey of more than 1,300 business and IT decision makers, 60% of respondents reported that half or more of their organization's data is considered dark. A full one-third of respondents reported this amount to be 75% or more.[1]

Source: https://www.ibm.com/think/topics/dark-data

Key big data statistics for 2026

- **Is big data still a strong career bet?**
  - Yes, big data and data science remain incredibly strong career bets. The US Bureau of Labor Statistics (BLS) projects a massive 34% growth for data scientists through 2034 with a median wage of $112,590. Furthermore, the G2 Big Data Statistics indicate the market is on pace to hit $1.18 trillion by 2034
  - **Global Market Expansion:** It isn't just a US phenomenon. The World Economic Forum ranks Big Data Specialists as the #1 fastest-growing global occupation, with strong demand hubs in the UK, Europe, and Asia. (https://www.ie.edu/uncover-ie/demand-for-data-science-professionals-master-in-business-analytics-and-data-science/)

What is big data? Is big data a new wave?

No & Yes

- No, most of these problems have been in the focus of data management research for years
- Yes, it is a different type of data wave:

  One needs to put together many sources of information coming through many different channels, throwing away what is not important, working under time constraints, serving analysts and end users

What is big data?

Most commonly accepted definition, by Gartner (3Vs)

"Big data is high-volume, high-velocity and high-variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making."

The amount of data generated is vast compared to traditional data sources

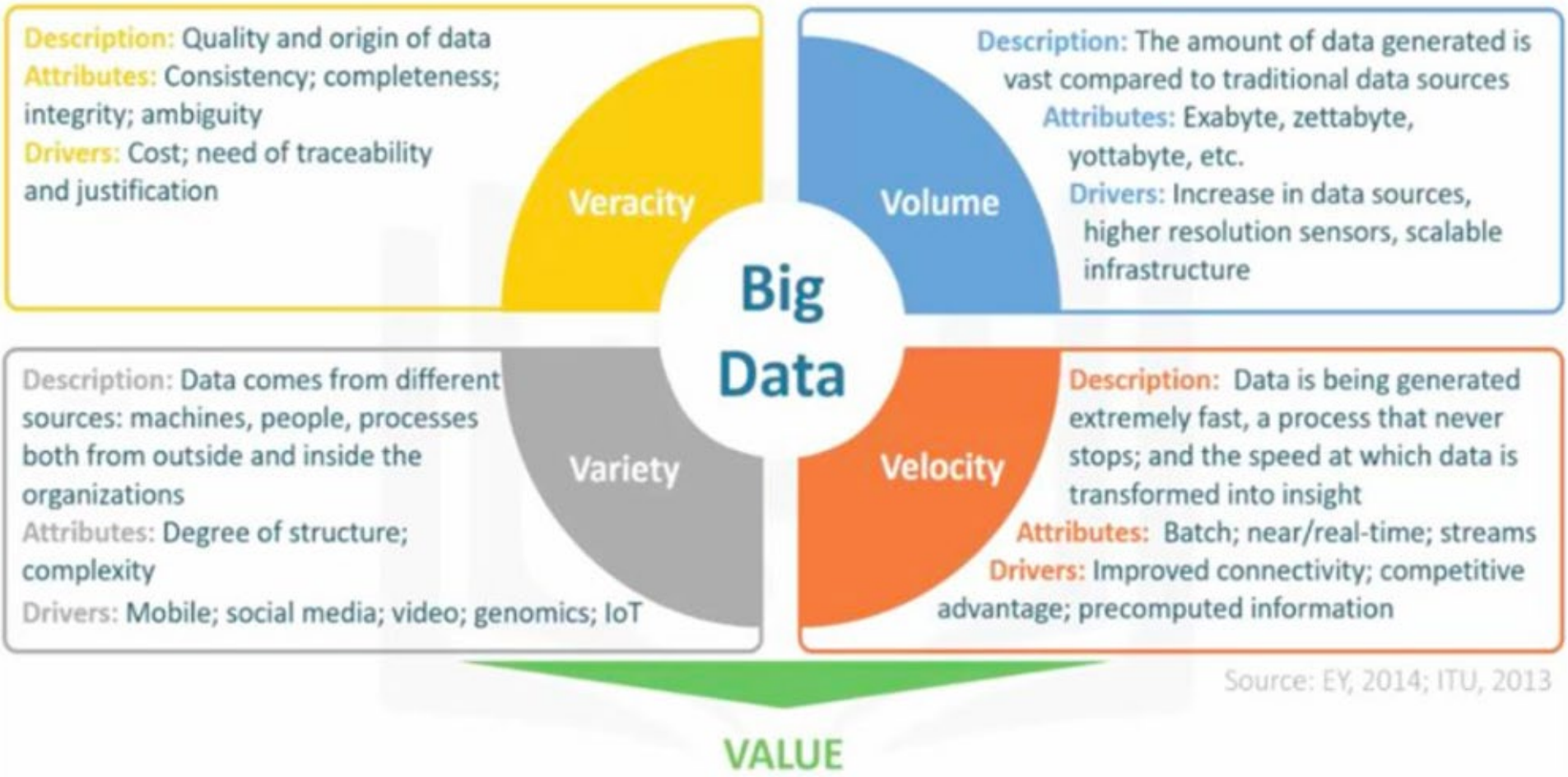
- exabyte, zettabyte, yottabyte, etc
- More data sources, higher resolution sensors

- Data is being generated extremely fast,
- a process never stops
- the speed at which data is transformed into insight.

Data comes from different sources

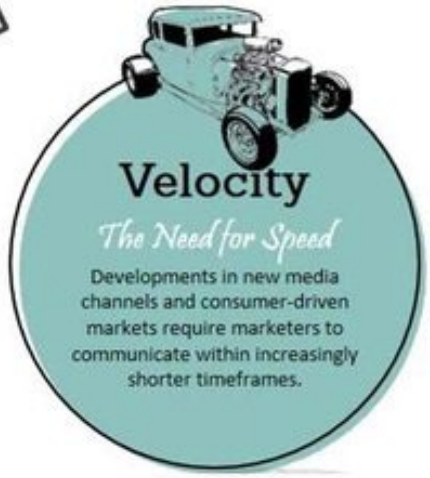
- numerical, textual, spatial and temporal, videos, images, transactions, graph, etc
- Temperatures, traffic speed, population distribution, electricity consumption, social media

More than 3Vs

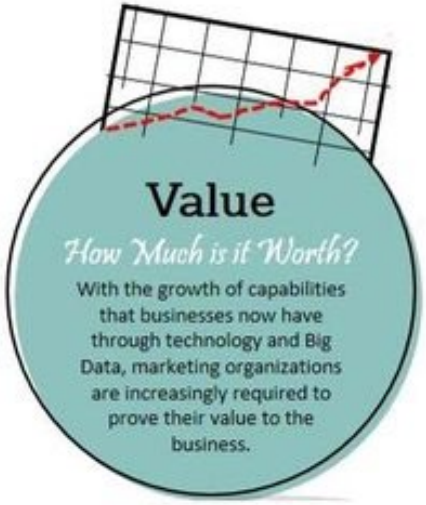


More than 3Vs – some make it 4Vs

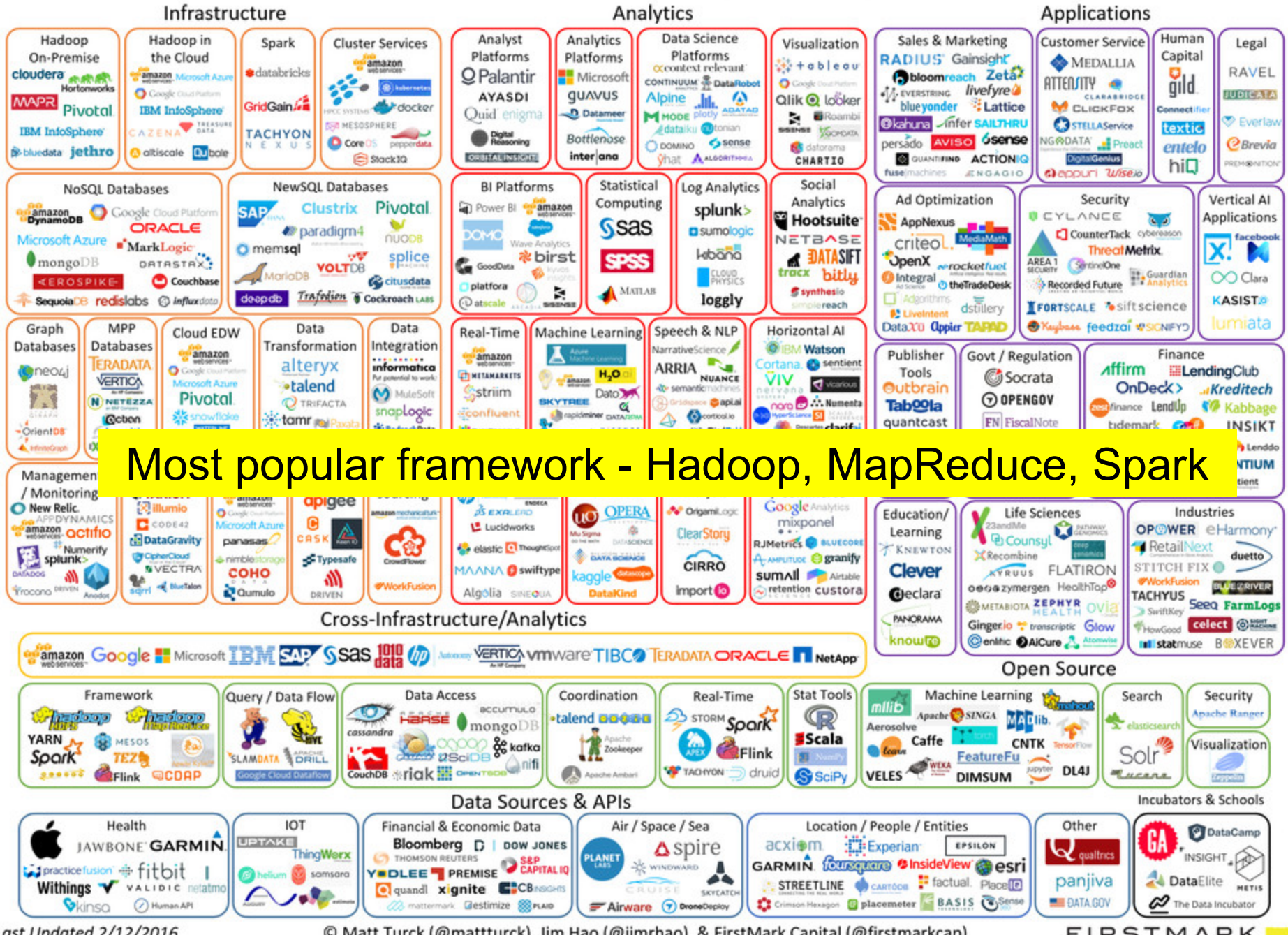








Big Data Landscape 2016 (Version 2.0)



Big Data - Big Ideas

Tackling large-data problems requires a distinct approach

- Scale “out”, not “up”
- Failure are common
- Move processing to the data

Big Data - Big Ideas

Scale “out”, not “up”

The data-intensive workloads today are beyond the capability of any single machine, no matter how powerful

- Two Approaches
  - Using a large number of commodity low-end servers is the scaling “out ” approach
  - Using a small number of high-end servers is the scaling “up” approach

The scaling “out ” approach
is preferred over the scaling “up” approach.

Most existing implementations of the MapReduce programming model are designed around clusters of low-end commodity servers.

Big Data - Big Ideas

Tackling large-data problems requires a distinct approach

- Scale “out”, not “up”
- Failure are common
- Move processing to the data

RMIT Classification: Trusted

Big Ideas

Failure are common

Failures are common in a cluster of large number of low-end servers

- For a 10,000-server cluster,
  - The mean-time between failures (MTBF) of each server is 1000 days. There are 10 failures a day in average.

    What if MTBF is 2000 days?
- A well-designed, fault-tolerant service must ensure the expected quality of services
  - As servers go down, other cluster nodes should seamlessly take over the load.
  - A broken server that has been repaired should be able to seamlessly rejoin the service without manual reconfiguration by the administrator.

Big Data - Big Ideas

Tackling large-data problems requires a distinct approach

- Scale “out”, not “up”
- Failure are common
- Move processing to the data

Big Data - Big Ideas

Move processing to the data

In traditional high-performance computing (HPC) applications (e.g., climate or nuclear simulations), *processing nodes* and *storage nodes* linked together by a high-capacity interconnect.

- Many data-intensive workloads are not very processor-demanding (e.g., word count, time complexity is linear).
  - Any general processor such as the one in storage node can be used
  - Moving large volume of data becomes the performance bottleneck
    - processor wait for data

So, moving and running the processing code to the nodes where data are stored.

End of Part 2

Part 3
Course overview

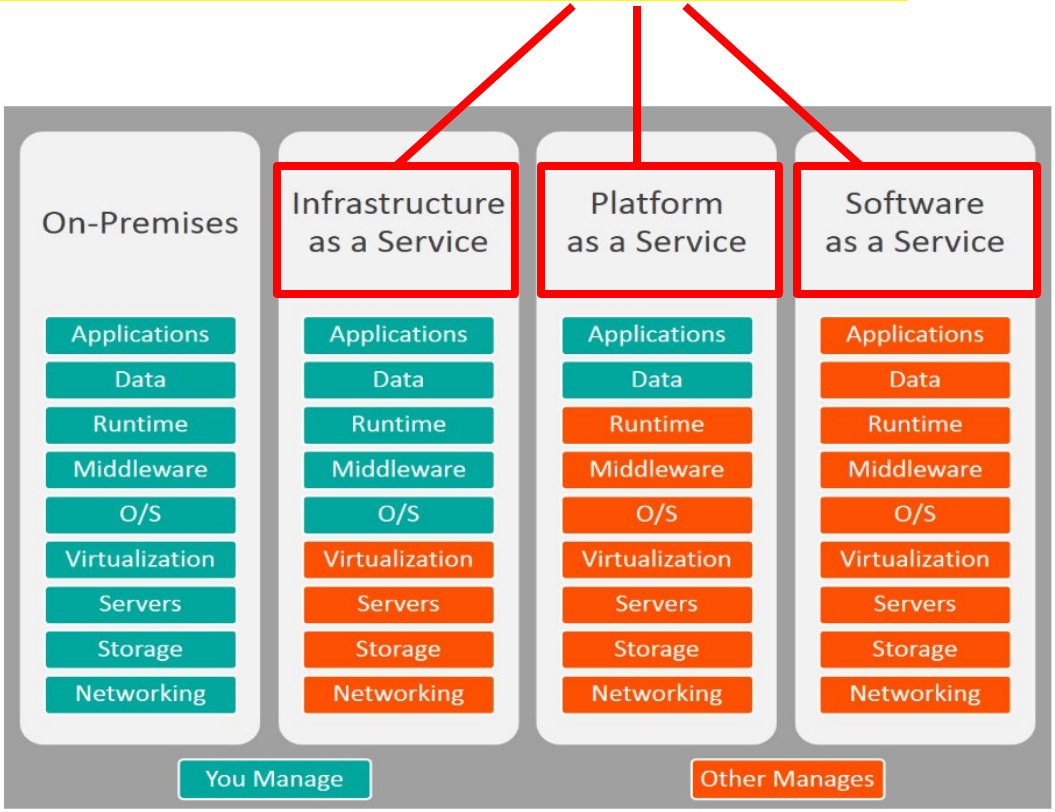
Course overview

Week 1 – Introduction to big data processing
Week 2 – Hadoop, HDFS, YARN
Week 3 – MapReduce Basics
Week 5 – MapReduce Algorithm Design Patterns
Week 4 – Advanced MapReduce Programming - Clustering
Week 6 – Advanced MapReduce Programming on Graph
Week 7 – PIG, HIVE, Mahout, Solr, Storm (1)
Week 8 – PIG, HIVE, Mahout, Solr, Storm (2)
Week 9 – Spark Basics
Week 10 – Spark streaming 1
Week 11 – Spark streaming 2
Week 12 – Course Revision

Course overview

Week 2 - Hadoop, HDFS, YARN

- Data Center and Cloud computing
- Utility Computing Everything as a Service



Course overview

Week 2 - Hadoop, HDFS, YARN

- Apache Hadoop Ecosystem
  - Hadoop Distributed File System (HDFS)

A distributed file-system that stores data on commodity machines, providing very high aggregate bandwidth across the cluster;

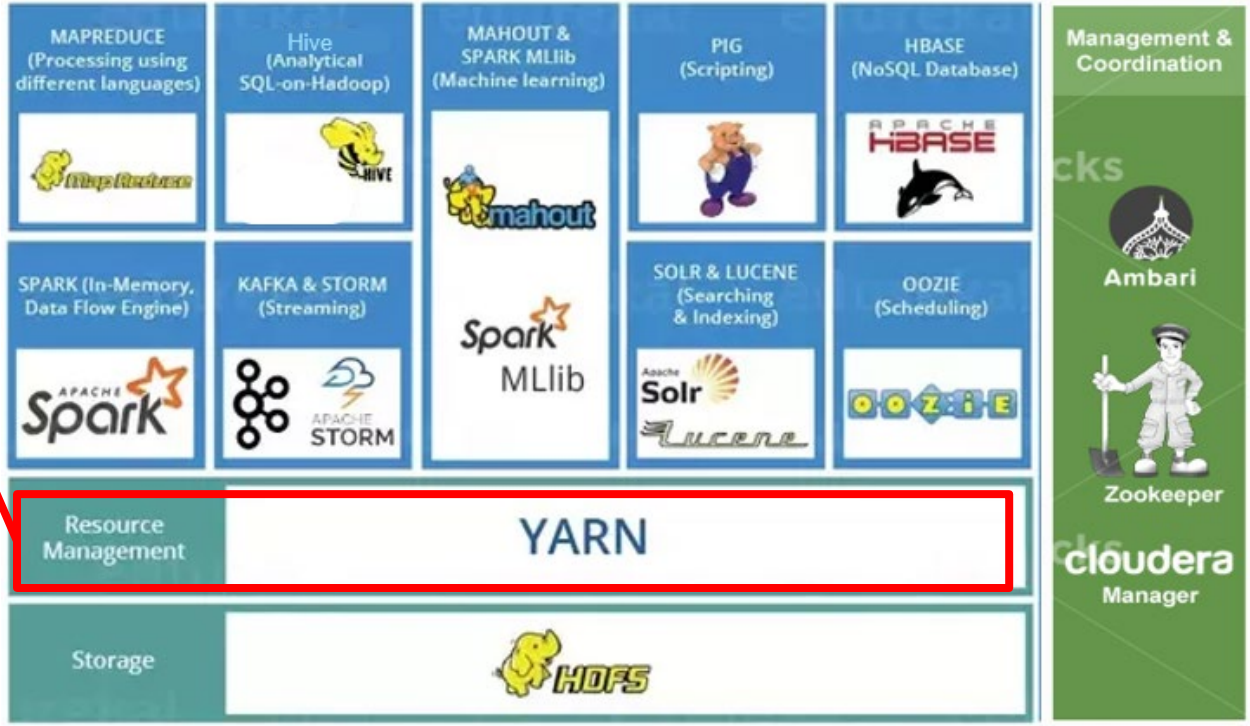


Course overview

Week 2 - Hadoop, HDFS, YARN

- Apache Hadoop Ecosystem
  - Yet Another Resource Negotiator (Yarn)

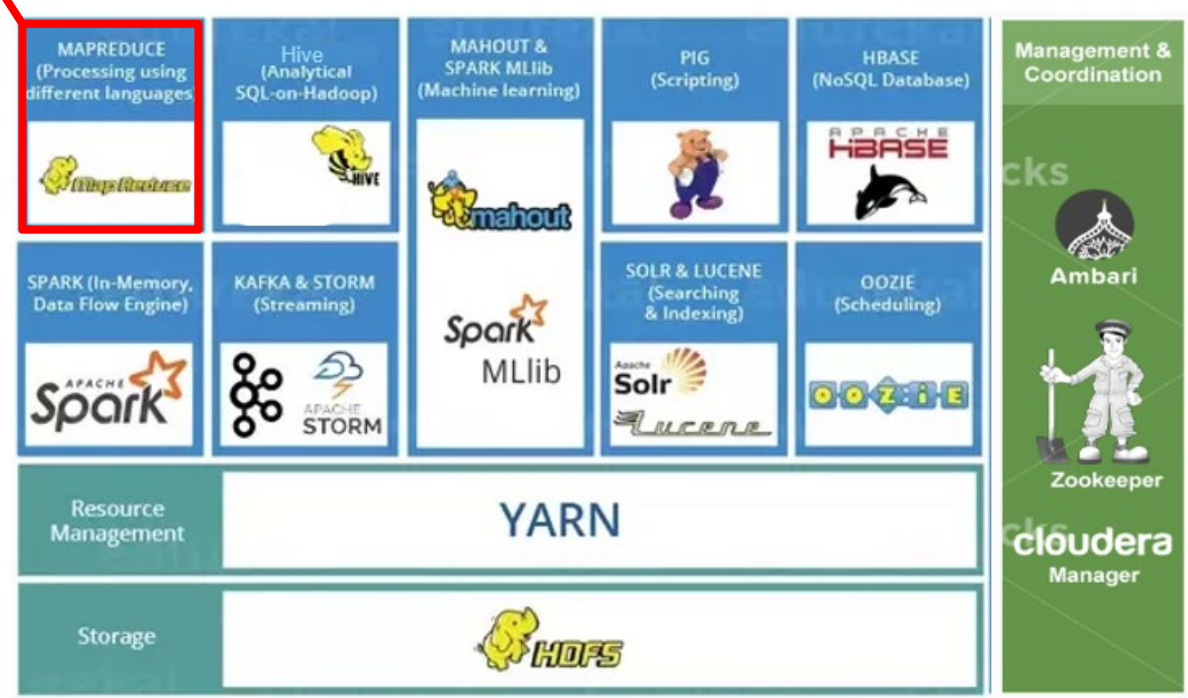
A platform responsible for managing computing resources in clusters and using them for scheduling users' applications



Course overview

Week 3 - MapReduce Basics

MapReduce is programming model for large-scale data processing.



Course overview

Week 3 - MapReduce Basics

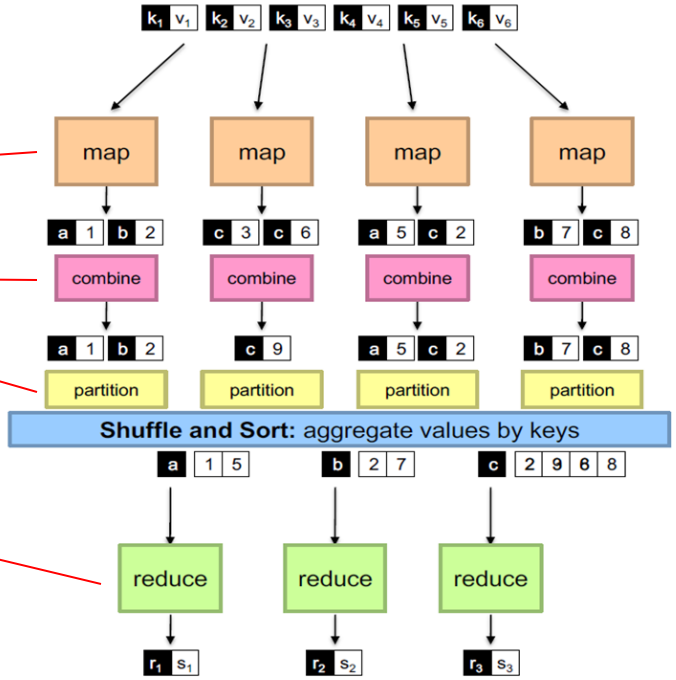
- Programmers just specify four functions:

map $(k_1, v_1) \rightarrow [<k_2, v_2>]$

combine $(k_2, [v_2]) \rightarrow [<k_2, v_2>]$

partition (k', number of partitions) → partition for k'

reduce $(k_2, [v_2]) \rightarrow [<k_3, v_3>]$



Course overview

Week 3 - MapReduce Basics

- Programmers just specify four functions:

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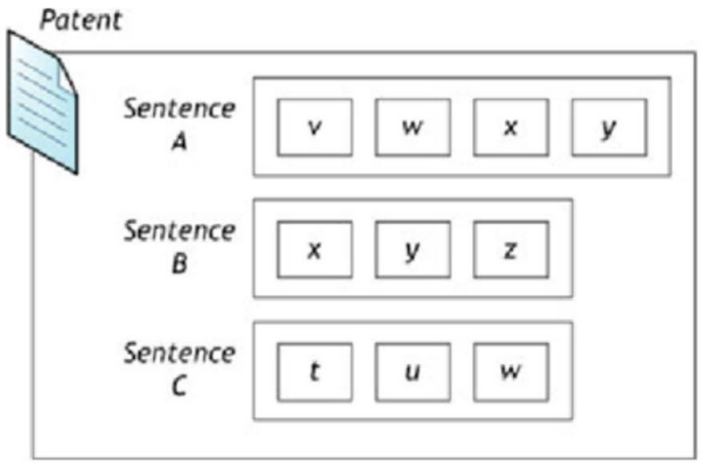
All other aspects of execution are handled transparently by the execution framework on clusters

Course overview

Week 4 - MapReduce Algorithm Design Patterns via building term co-occurrence matrix

- MapReduce Algorithms for term co-occurrence matrix for a text collection

Term co-occurrence matrix – example



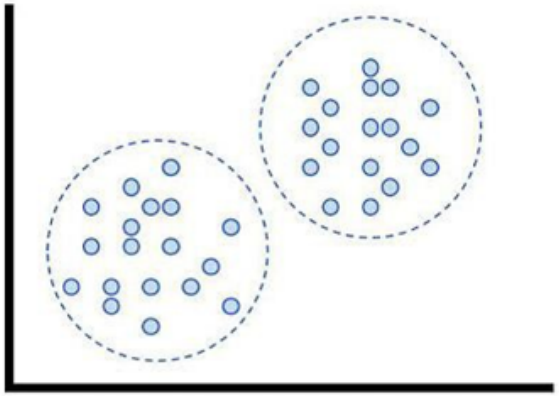
Co-occurrence matrix

	v	w	x	y	z	t	u
v	0	1	1	1	0	0	0
w	1	0	1	1	0	1	1
x	1	1	0	2	1	0	0
y	1	1	2	0	1	0	0
z	0	0	1	1	0	0	0
t	0	1	0	0	0	0	1
u	0	1	0	0	0	1	0

Course overview

Week 5 - MapReduce Programming and Clustering

Clustering
Unsupervised
Unknown number of classes
No prior knowledge
Used to understand (explore) data



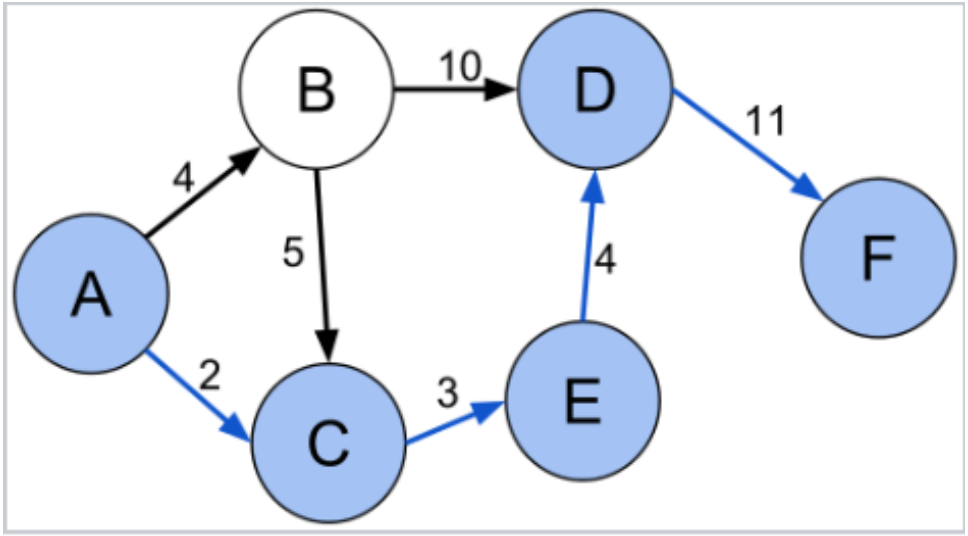
Clustering

Investigate implementation detail for MapReduce K-means clustering

Course overview

Week 6 – MapReduce Large Graph Processing

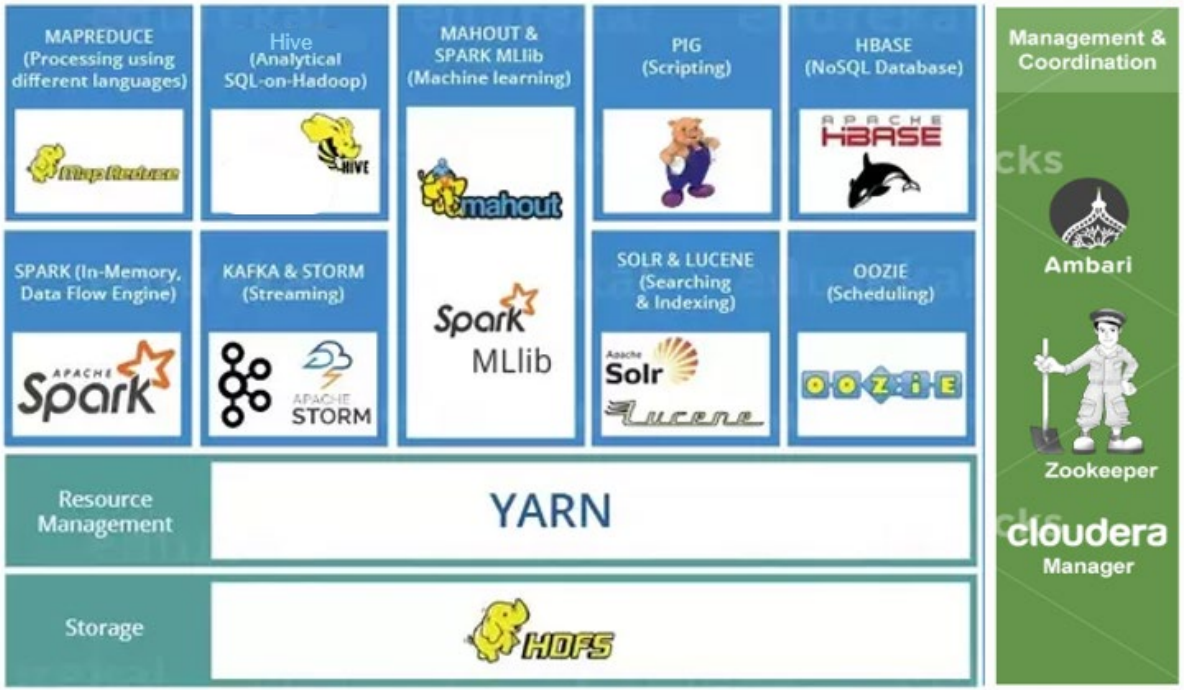
- road networks, online social media, utility networks, WWW …
- Shortest Path Problem
  - find shortest path from a source node to one or more target nodes



From A to F

Course overview

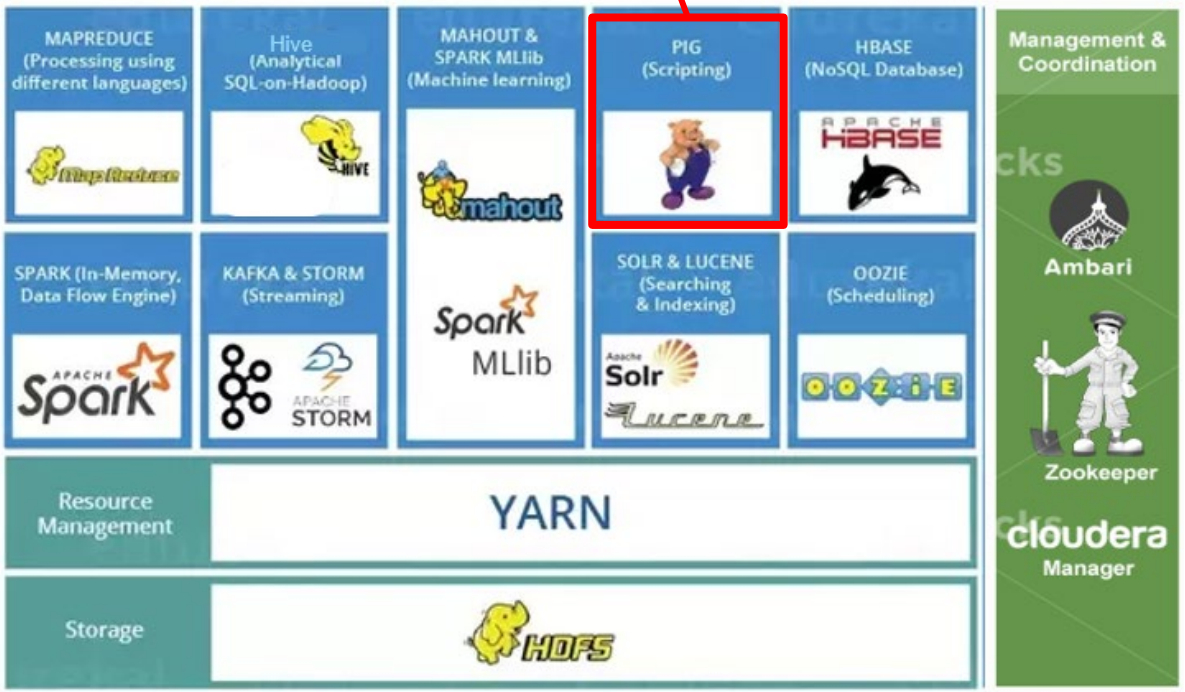
Week 7, 8 - PIG, HIVE, Mahout, Solr, Storm



Course overview

Week 7 - PIG, HIVE, Mahout, Solr, Storm (1)

- Pig: large-scale data processing system
  - Scripts are written in Pig Latin, a dataflow language
  - Programmer focuses on data transformations
  - Developed by Yahoo!, now open source



Course overview

Week 7 - PIG, HIVE, Mahout, Solr, Storm (1)

- Hive: data warehousing application in Hadoop
  - Query language is HQL, variant of SQL
  - Tables stored on HDFS with different encodings
  - Developed by Facebook, now open source



Course overview

Week 8 - PIG, HIVE, Mahout, Solr, Storm (2)

- Mahout offers a ready-to-use framework for doing data mining tasks on large volumes of data,
  - For example, k-means, fuzzy k-means, Canopy, Dirichlet, and Mean-Shift, Naive Bayes and Complementary Naive Bayes classification



Course overview

Week 8 - PIG, HIVE, Mahout, Solr, Storm (2)

- Solr runs as a standalone full-text search server by providing distributed indexing, replication and load-balanced querying,
- Solr is highly reliable, scalable and fault tolerant, automated failover and recovery, centralized configuration and more.



Course overview

Week 8 - PIG, HIVE, Mahout, Solr, Storm (2)

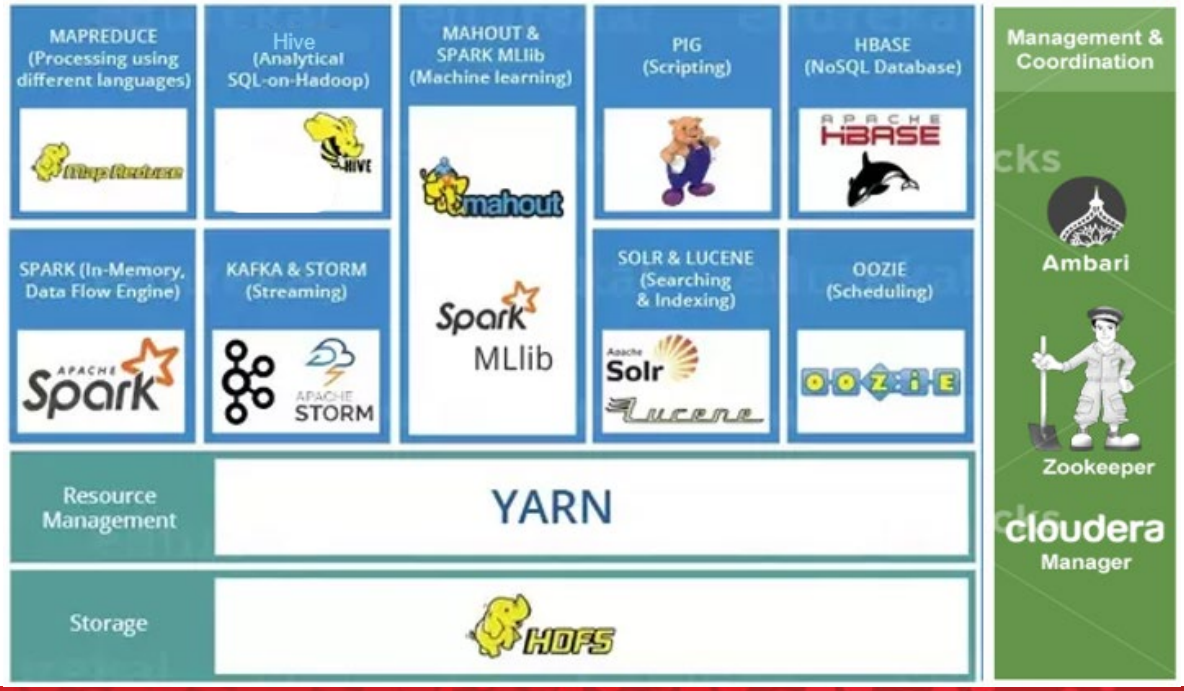
- Apache Storm is a free and open-source distributed real-time computation system
- Apache Storm makes it easy to reliably process unbounded streams of data, doing for real-time processing what Hadoop did for batch processing.



Course overview

Week 9 - Spark Basics

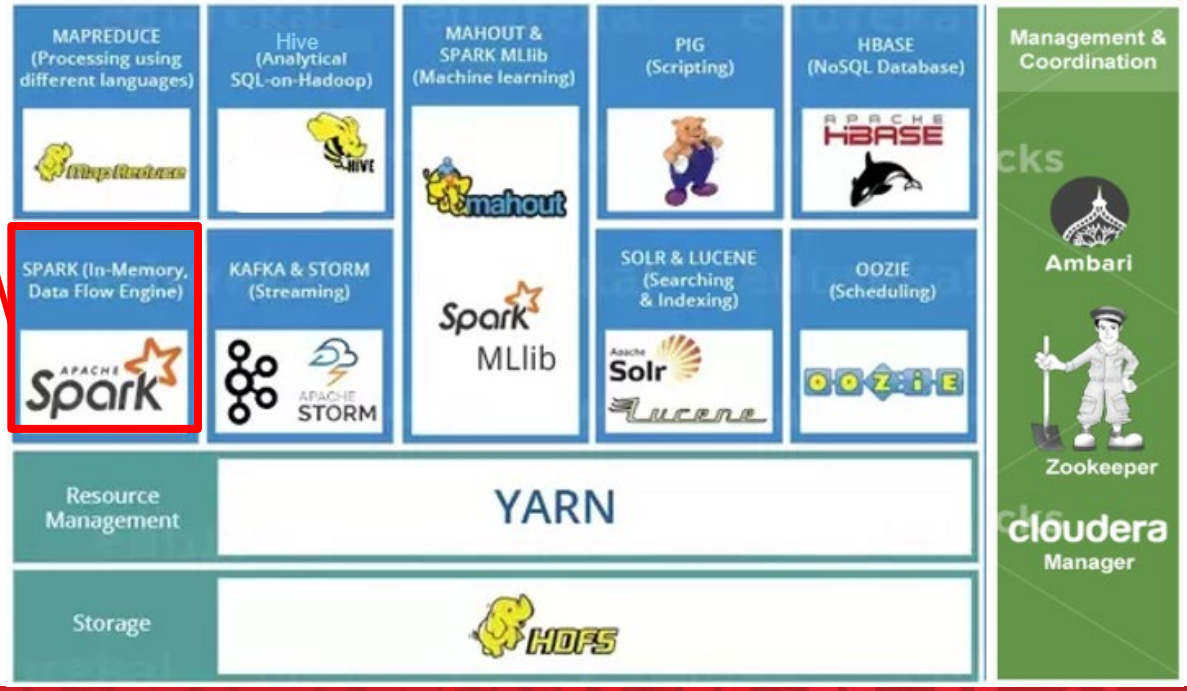
An open-source and fast engine for large-scale data processing.



Course overview

Week 9 - Spark Basics

An open-source and fast engine for large-scale data processing.

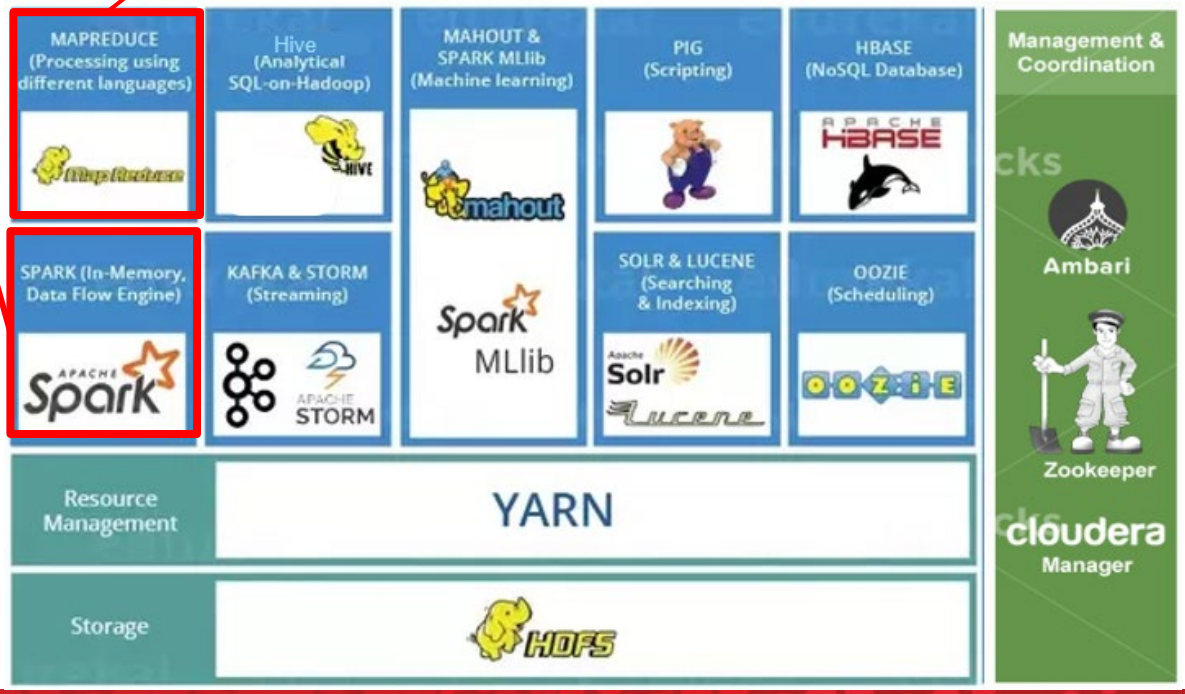


Course overview

Week 9 - Spark Basics

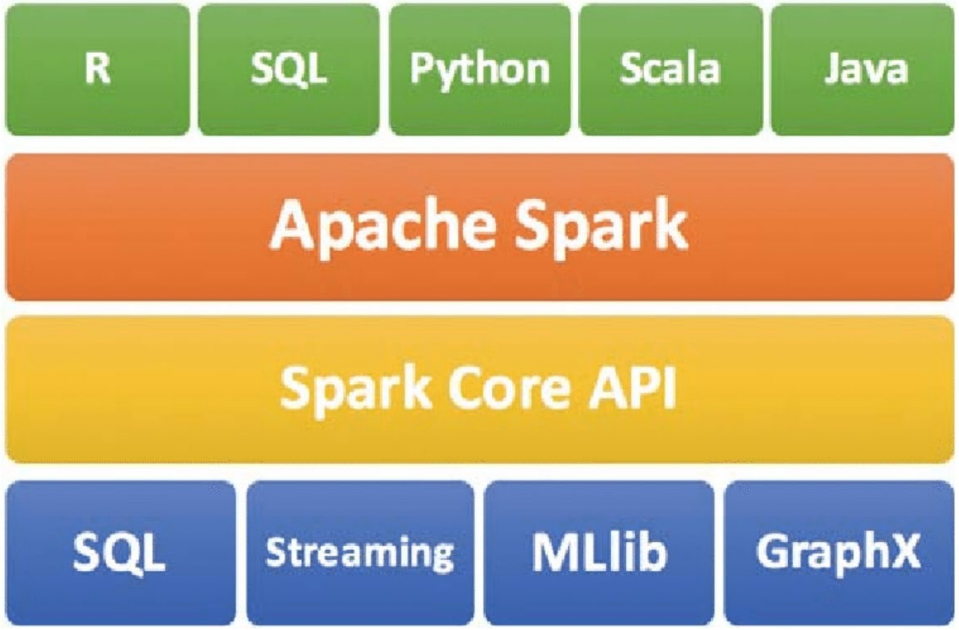
An open-source and fast engine for large-scale data processing.

- Apache Spark vs. Apache Hadoop MapReduce
  - Spark is generally a lot faster than MapReduce because of the way it processes data.



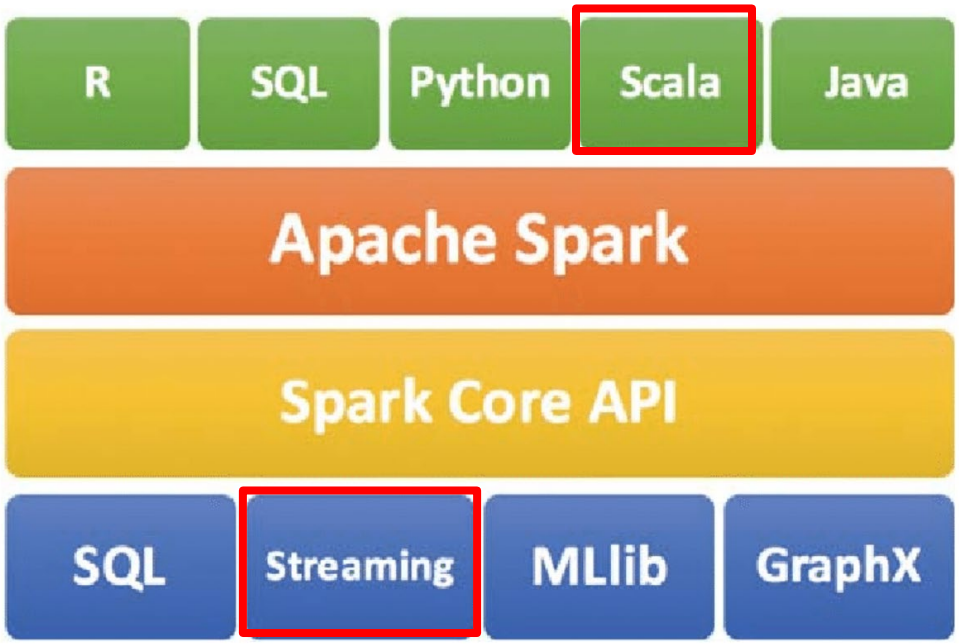
Course overview

Week 9 - Spark Basics



Course overview

Week 9 - Spark Basics

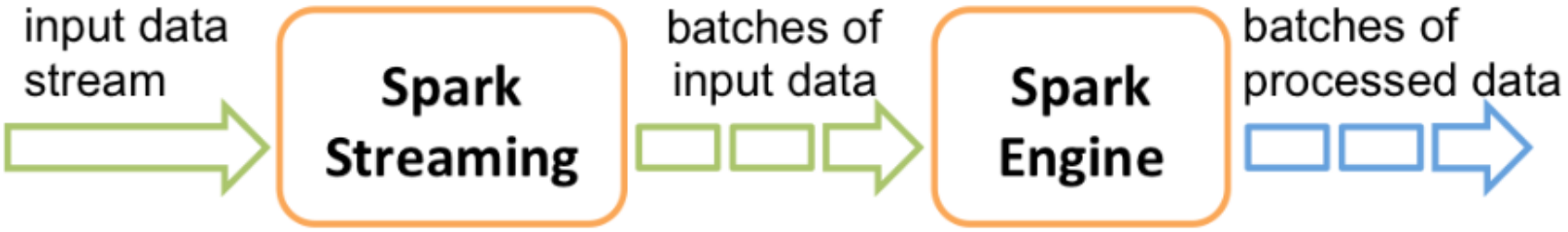


Spark supports SQL, data streaming machine learning and graph processing.

Course overview

Week 10, 11 – Spark streaming

- Spark Streaming is an extension of the core Spark API that enables scalable, high-throughput, fault-tolerant stream processing of live data streams.



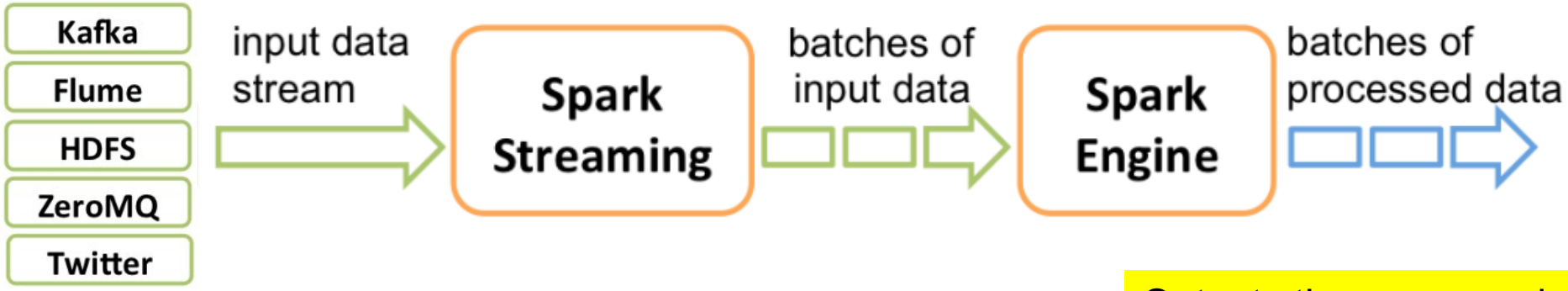
Spark Streaming receives live input data streams

Divides the data into batches

Then process each batch by the Spark engine to generate the final stream of results in batches

Course overview

Week 10, 11 – Spark streaming



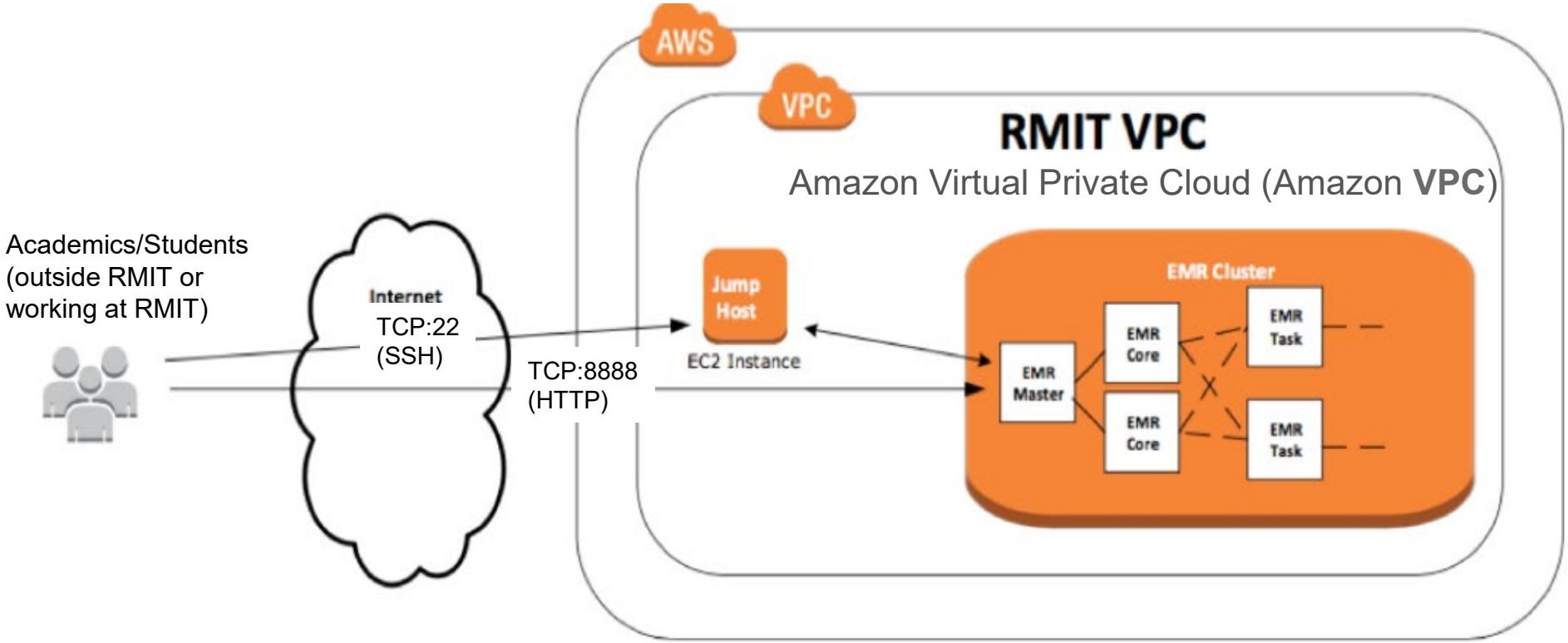
Inputs sources can be
- Flume,
- HDFS,
- KAFKA,
- etc

Outputs the processed data to
- filesystems
- database
- live dashboards

AWS Environment in Lab Classes

The labs use AWS EMR (Amazon Elastic MapReduce)

- We have created an JumpHost account for each student,
  - The access key will send to you this week
- The first lab instruct details how to access JumpHost and then access EMR



Assessment

- **25% Assignment 1**
- **25% Assignment 2**
- **25% Assignment 3**
- **25% Timed Assessment**
  - This is a 2-hour assessment that may be taken at any time within a 24-hour period.
- **No final exam**

Academic Integrity

University policy - students caught passing off somebody else's work are subject to discipline policy

- There is a difference between copying somebody else's text and summarizing their content in your own words after crediting the source
- We will check assignments for plagiarism!

http://www.rmit.edu.au/students/academic-integrity

End of Part 3

Next Week
Hadoop, HDFS, YARN